

Wearable Automatic ASSISTANCE GLOVE

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This proposed prototype works by wearing the glove and either touching a specific finger with the thumb or pressing a special switch. This action triggers a bell (buzzer) to ring automatically for 5 seconds, alerting others that assistance is needed. Additionally, the name of the requested item is displayed on the LCD screen for 10 seconds. This device has applications across various fields, including hospitals, offices, and even restaurants.

In a hospital, a caretaker assists patients with daily tasks, such as administering medicine, feeding meals, and walking. Often, an elderly patient may struggle to communicate their needs due to difficulty in expressing themselves due to illness or weakness.

LIST OF COMPONENTS

Components	Quantity
433MHz Tx-Rx module (RF1, RF2)	1
Decoder IC HT12D (IC1)	1
Encoder IC HT12E (IC2)	1
Arduino Uno (MOD 1)	1
12V, 1A adaptor	1
Buzzer (B1)	1
5mm blue LED (D1, D2, D3, D4)	4
Push to on switch (S1, S2, S3, S4)	4
3.7V rechargeable battery	2
3.7V rechargeable battery case/holder	1
16x2 LCD (MOD 2)	1
5V voltage regulator IC7805 (IC3)	1
Resistors 1-mega-ohm (R1), 47-kilo-ohm (R2, R3)	3
10-kilo-ohm potmeter (POT 1)	1

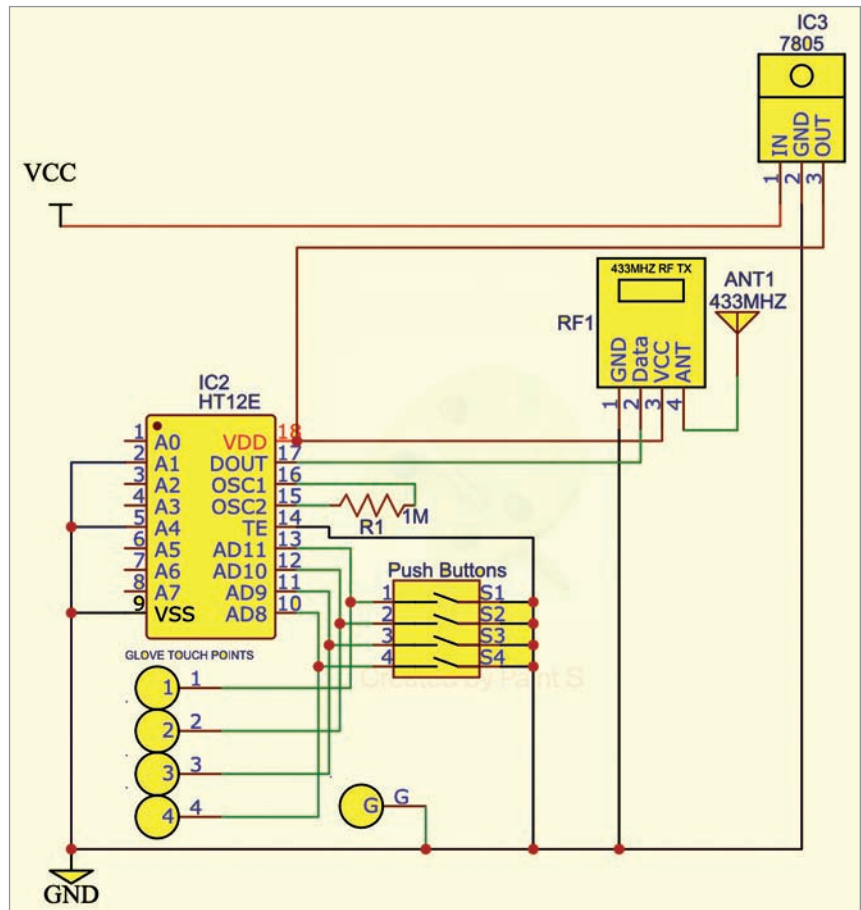


Fig. 1: Transmitter circuit

This device solves that issue by allowing patients to notify caretakers easily.

For mute patients, communication challenges can be even greater. In such cases, this device provides an effective solution, enabling patients to communicate their needs without verbal expression.

In office settings, during general meetings, ordering refreshments like tea, coffee, or snacks usually

requires ringing a bell and waiting for an attendant. This process can cause delays and interruptions. With this device, orders can be placed quickly and discreetly, avoiding disruptions during meetings and saving time.

In restaurants, the device proves equally useful. With high customer volumes, waiters may struggle to serve customers promptly. When installed at tables, this system can